

Small Matrices - Run Instructions

A step-by-step guide for a reader with no previous programming experience

Purpose of this document. This guide explains exactly how to obtain the Python files, prepare a simple Anaconda environment, run the benchmark, find the CSV results, and generate and locate the plots. It is deliberately written so that a reader who has never used Python can follow it from beginning to end.

Important: this is the standalone local version of the experiment. It is not a Slurm job and it does not require access to the University of Manchester CSF or any other high-performance computing cluster.

1. What should be in the folder

The Small Matrices folder should contain the following five repository files:

File	Purpose
Small_Matrices.py	Runs the numerical benchmark and creates the CSV result files.
Plot_Small_Matrices.py	Reads the benchmark results and creates the PNG figures.
Small_Matrices_Introduction.pdf	Plain-English explanation of what the experiment is testing.
Small_Matrices_Run_Instructions.pdf	This step-by-step reproduction guide.
Small_Matrices_Requirements_and_Restrictions.pdf	Software requirements, assumptions and limitations.

Generated files are different. The CSV files and PNG figures are outputs produced by the scripts. They do not need to be stored in the GitHub repository. They can be generated again whenever the experiment needs to be reproduced.

2. What you need before starting

You need a normal computer, an internet connection for the initial software installation, and either Anaconda or Miniconda. No knowledge of programming is assumed.

Before running anything, read **Small_Matrices_Requirements_and_Restrictions.pdf**. It lists the Python packages and the limitations that apply to interpreting the results.

The benchmark uses standard scientific Python libraries. The intended local workflow is: create the environment, open the folder in a terminal, run the benchmark, inspect the CSV files, then run the plotting script.

3. Step 1 - Install Anaconda or Miniconda

If Anaconda or Miniconda is already installed, skip to Section 4. If it is not installed, install one of them using the normal installer for your operating system. After installation, open a terminal. On macOS or Linux this is the Terminal application. On Windows, use Anaconda Prompt or a terminal where the **python** command is available.

Do not open the Python files by double-clicking them. They are intended to be run from a terminal so that you can see progress messages and any error messages.

4. Step 2 - Open the Small Matrices folder in a terminal

The terminal must be working inside the folder containing the two Python files. The easiest method is to open the folder in Finder/File Explorer, then use the terminal's directory-changing command.

macOS or Linux: type **cd**, followed by a space, then drag the Small Matrices folder into the terminal window. Press Enter.

For example, the command will look similar to:

```
cd /path/to/Small\ Matrices
```

The exact path on your computer will be different. You do not need to type the example path literally.

Check that you are in the correct folder by listing its contents. On macOS/Linux use:

```
ls
```

You should see **Small_Matrices.py** and **Plot_Small_Matrices.py** in the list. If you do not, stop here and change to the correct folder.

5. Step 3 - Create the Python environment

The exact package list should be taken from the companion Requirements and Restrictions document. If that document provides a ready-made environment command or environment file, follow that instruction first. The purpose of this step is simply to make sure that Python, NumPy, SciPy, pandas and Matplotlib are available.

A typical check is:

```
python --version
```

Then check the main scientific packages:

```
python -c "import numpy, scipy, pandas, matplotlib; print('Required packages are available.')" 
```

If this command prints the confirmation message, the basic environment is ready. If it reports that a package is missing, install the missing package in the environment before continuing.

6. Step 4 - Run the benchmark

Make sure the terminal is still inside the Small Matrices folder. Run:

```
python Small_Matrices.py
```

The program creates the synthetic matrices in memory and evaluates four solvers: Direct, CG, LSQR and LSMR. It changes the specified experimental factors while keeping the other factors at their default values. Each selected configuration is repeated ten times.

While the program is running, you should see progress information in the terminal. Do not close the terminal. The benchmark may take a substantial amount of time depending on the computer.

When the benchmark finishes successfully, it writes the numerical results to the **output_small_matrices** directory. The main result file is **small_matrix_results.csv**. Summary CSV files are also created.

7. The benchmark settings

Parameter	Recorded setting
Matrix rows (m)	400, 500, 1000, 1100, 2100
Matrix columns (n)	50 fixed for the small-matrix section
Condition parameter	10, 100, 1000, 10000
Regularisation lambda	0, 1e-8, 1e-6, 1e-4, 1e-2, 1e-1, 1
Sparsity	0, 0.25, 0.5, 0.75, 0.9
Sparsity patterns	random, banded, block
Spectra	clustered, exponential, linear, polynomial
Repeats	10 per configuration
Relative noise	0.01
Tolerance	1e-8

Parameter	Recorded setting
Maximum iterations	5000
Random seed	42

These values describe the intended dissertation experiment. In particular, the regularisation values are written here in plain ASCII notation such as **1e-8** rather than using special mathematical characters. This avoids the black-block character problem that can occur in some PDF viewers.

8. Step 5 - Find the CSV results

After the benchmark has finished, return to the Small Matrices folder using Finder/File Explorer. A new directory called **output_small_matrices** should have been created if the scripts use their default output location.

Open that directory. The principal numerical output is:

```
output_small_matrices/small_matrix_results.csv
```

This file contains the recorded solver-level results. It can be opened with Excel, LibreOffice Calc, Numbers or another spreadsheet program. It is also a normal CSV text file, so it can be inspected with a text editor.

The benchmark also produces summary CSV files. The standard set includes:

CSV file	What it contains
small_matrix_results.csv	Raw recorded results for the solver runs.
small_matrix_summary.csv	Aggregated results across repeated runs.
matrix_size_summary.csv	Summary for the matrix-size experiment.
condition_number_summary.csv	Summary for the condition-parameter experiment.
lambda_summary.csv	Summary for the regularisation experiment.
sparsity_summary.csv	Summary for the sparsity experiment.
pattern_summary.csv	Summary for the sparsity-pattern experiment.
spectrum_summary.csv	Summary for the spectral-distribution experiment.
small_matrix_metadata.txt	A text record of the run configuration.

If a particular file is not present, do not assume the benchmark failed. The exact files depend on the version of the benchmark script. First check the terminal output and the output directory.

9. Step 6 - Generate the plots

Do this **only after** the benchmark has successfully produced its CSV results. Return to the terminal and make sure you are still in the Small Matrices folder.

Run:

```
python Plot_Small_Matrices.py
```

The plotting script reads the benchmark results and creates PNG figures in:

```
output_small_matrices/plots/
```

The figures compare the four solvers using measures such as runtime, iteration count, residual or reconstruction error, and distributions of measured performance. The plotting script is a post-processing step: it does not create the benchmark data itself.

10. How to open and save the plots

Open the **output_small_matrices** folder in Finder/File Explorer and then open the **plots** subfolder. The generated PNG files can be opened normally with the computer's image viewer.

If you need a plot for a dissertation figure, copy the required PNG file to another folder or insert it directly into your document. The figures are generated at high resolution by the plotting script.

You do not need to move the PNG files back into the repository. They are derived outputs and can be regenerated from the Python source files.

11. What to do if the plotting script says that a CSV is missing

This is the most common problem when the two scripts are run in the wrong order. First confirm that **Small_Matrices.py** has finished without an error.

Then confirm that **output_small_matrices** exists and contains **small_matrix_results.csv**.

Finally, confirm that both Python files are being run from the same Small Matrices folder and that neither script has been edited to use a different output directory.

If the result CSV exists but has a different name, the plotting script must be configured to read that exact filename. Do not rename the CSV blindly, because the plotting code may refer to the original name.

12. A complete start-to-finish example

The following is the shortest complete workflow. A beginner can use it as a checklist after reading the detailed sections above.

Step A: Open a terminal.

Step B: Change directory into the Small Matrices folder.

```
cd /path/to/Small\ Matrices
```

Step C: Check that the two Python files are present.

```
ls
```

Step D: Check Python and the required packages.

```
python --version
```

```
python -c "import numpy, scipy, pandas, matplotlib; print('Required packages are available.')" 
```

Step E: Run the numerical benchmark.

```
python Small_Matrices.py
```

Step F: Wait until the benchmark reports completion. Then look for **output_small_matrices/small_matrix_results.csv**.

Step G: Generate the plots.

```
python Plot_Small_Matrices.py
```

Step H: Open **output_small_matrices/plots/** to view the PNG figures.

Step I: If you need the numerical data, copy the CSV files from **output_small_matrices/** to another location or open them in a spreadsheet program.

13. What should remain in the GitHub repository?

For a clean reproducibility repository, keep the two Python source files and the three explanatory PDFs. The generated CSV and PNG files can be removed after you have verified that the scripts run correctly.

A suitable Small Matrices folder therefore contains:

```
Small_Matrices.py Plot_Small_Matrices.py Small_Matrices_Introduction.pdf  
Small_Matrices_Run_Instructions.pdf Small_Matrices_Requirements_and_Restrictions.pdf
```

Do not commit temporary Python cache folders, notebook checkpoint directories, operating-system metadata, or other files created only by the local computer.

14. Reproducibility notes

The fixed random seed makes the synthetic problem construction reproducible for the recorded implementation. It does **not** mean that every computer will report exactly the same runtime. Timing depends on the CPU, BLAS/LAPACK implementation, Python and scientific-library versions, operating-system activity, and other background processes.

The condition values 10, 100, 1000 and 10000 are **condition parameters used to control spectral scaling**. They should not automatically be interpreted as the exact condition number of every final sparse matrix, because sparsification can change the realised spectrum.

The benchmark uses a known synthetic reference solution, which allows reconstruction error to be measured as well as solver convergence. The added relative noise is 0.01.

15. If you are completely new to Python

You do not need to understand the Python code in order to reproduce the experiment. Think of **Small_Matrices.py** as the experiment itself and **Plot_Small_Matrices.py** as the program that turns the experiment's numerical results into graphs.

The important distinction is:

You run...	It produces...
Small_Matrices.py	Numerical benchmark results, CSV summaries and metadata.
Plot_Small_Matrices.py	PNG figures based on the CSV results.

If you can follow the commands in Section 12, you can reproduce the computational experiment without writing or modifying any Python yourself.

End of Small Matrices Run Instructions